ME1100
Thermodynamics

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Class timing

1. **Monday: 1 pm – 1:50 pm (G2 slot)**
2. **Tuesday: 2 pm -2:50 pm (C2 slot)**
3. **Thursday: 10 am – 10:50 am (G1 slot)**
4. **Friday: 9 am – 9:50 am (G1 slot)**

The quiz and end-semester will be as per G-slot timing

Course Contents

1. **Fundamentals: System & Control volume; Property, State & Process; Exact & Inexact differentials; Measurement of absolute pressure & Δp**
2. **Work: Thermodynamic definition of work; examples; Displacement work; Path dependence of displacement work and illustrations for simple processes; Fully resisted, partially resisted and unresisted process; Other forms of work - gravitational, electrical, magnetic, spring and shaft**
3. **Temperature & Heat: Definition of thermal equilibrium; Zeroth law; Definition of temperature and temperature scales; Various Thermometers; Definition of heat; Examples of heat/work interaction in systems**

Course Contents

4. **First Law for System: I Law for Cyclic & Non-cyclic processes; Concept of total energy E; Demonstration that E is a property; Various modes of energy**
5. **Pure substance: Two property rule; Enthalpy and internal energy; Properties of Ideal Gases and Mixtures of Ideal Gases; Properties of water-steam system; Const. temperature and Const. pressure heating of water; Definitions of saturated states; P-v-T surface; Use of steam and R134a tables- Saturation tables; Superheated tables, Identification of states & determination of properties**
6. **First Law for Flow Processes: Derivation of general energy equation for a control volume; Steady state steady flow processes; Examples of steady flow devices; Unsteady processes**

Course Contents

7. **Second law: Definitions of direct and reverse heat engines; Definitions of thermal efficiency and COP; Kelvin-Planck and Clausius statements; Definition of reversible process; Internal irreversibility and external irreversibility; Carnot cycle; Absolute temperature scale**
8. **Entropy: Clausius inequality; Definition of entropy S; Demonstration that entropy S is a property; Evaluation of ΔS for solids, liquids, and ideal gases undergoing various processes; Determination of s from steam tables; Examples - Turbine, compressor, pump, nozzle, diffuser; Definition of Isentropic efficiency; Available and Unavailable energy; Concept of Irreversibility and Lost work; Entropy balance of a control volume; Principle of increase of entropy; Entropy generation**

Books

1. **R.E. Sonntag, C. Borgnakke** and **G.J. Van Wylen,** 2003, 6th Edition, Fundamentals of Thermodynamics, John Wiley and Sons.
2. **Babu, V.**, Fundamentals of Engineering Thermodynamics, Second Edition, ANE Books Pvt. Ltd. 2019.
3. **Venkatesh, A**. Basic Engineering Thermodynamics, Universities Press (India) Limited, 2007.
4. **Moran, M. J. and Shapiro, H. N.**, 1999, Fundamentals of Engineering Thermodynamics, John Wiley and Sons.

Class Timing

DAY	1 8.00 – 8.50	2 9.00 – 9.50	3 10.00 - 10.50	4 11.00 – 11.50		5 13.00 - 13.50	6 14.00 - 14.50	7 15.00 – 15.50	8 16.00 –16.50	Extra 17.00 – 17.50
Mon	A1 EE1102 ESB 127	B1 MA1101 RJN 301	C1	D1 PH1010 RJN 301		G2 ME1100 RJN 301	B2	E2	C2	XX
		P2					P1 EE1102			
Tue	B1 MA1101 RJN 301	C1	D1 PH1010 RJN 301	E1		A2 EE1102 ESB 127 EE1103 ESB 242	C2 ME1100 RJN 301	D2	E2	YY
		Q2					Q1			
Wed	C1	D1 PH1010 RJN 301	E1	F1		B2 MA1101 RJN 301	D2	E2	F2	XX
		R2					R1 PH1030			
Thu	E1	F1	G1 ME1100 RJN 301	A1 EE1102 ESB 127 EE1103 ESB 242		D2 PH1010 (Tutorial) RJN 301	F2	G2	A2	YY MA1101 (Tutorial) RJN 301
		S2					S1 GN1101 FFT TERRACE			
Fri	F1	G1 ME1100 RJN 301	A1 EE1103 ESB 242	B1 EE1103 ESB 242		C2	F2	A2	B2	XX
		T2					T1			

Lunch recess 11.50 am –13.00 pm

Marks distribution

- **Quiz 1: 20%**
- **Quiz 2: 20%**
- **End-semester: 60%**

Introduction

Thermodynamics: **Therme +** **Dynamis**



Heat **Power**

Industrial revolution – Development of machines for performing labour-intensive tasks

- **Steam engines**
- **Internal combustion engines**

Conversion of heat to work

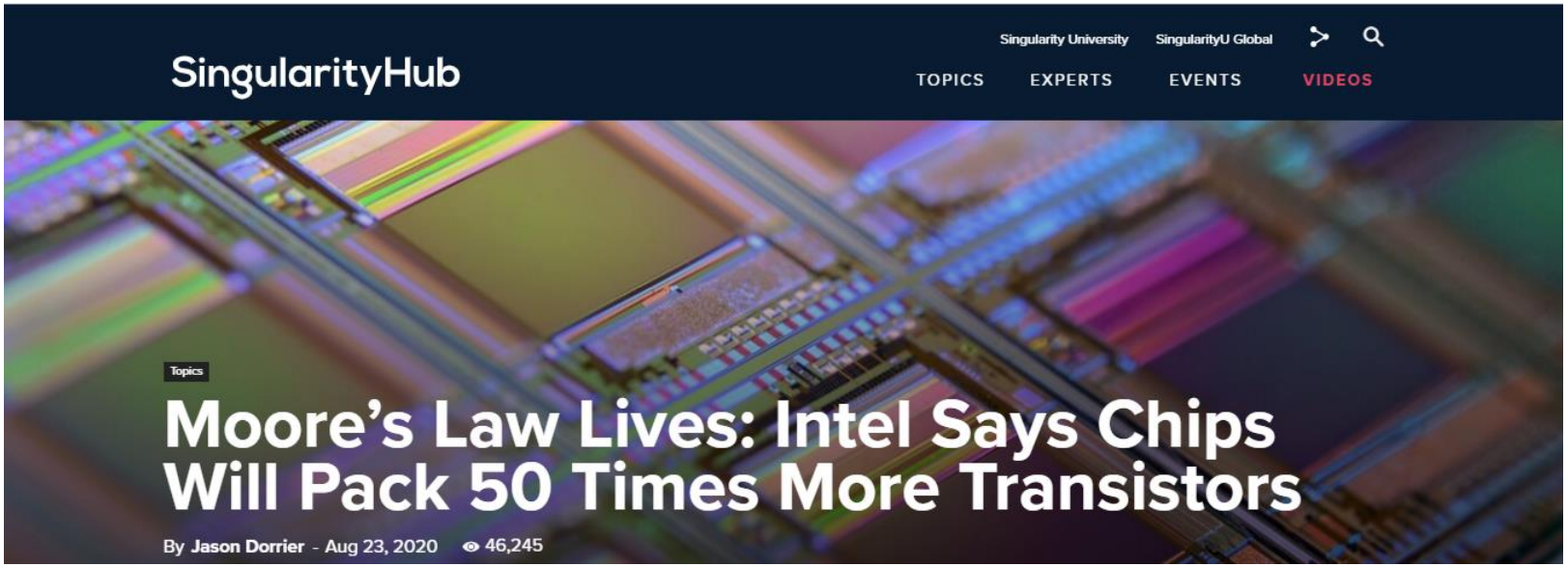
Realm of Mechanical Engineering

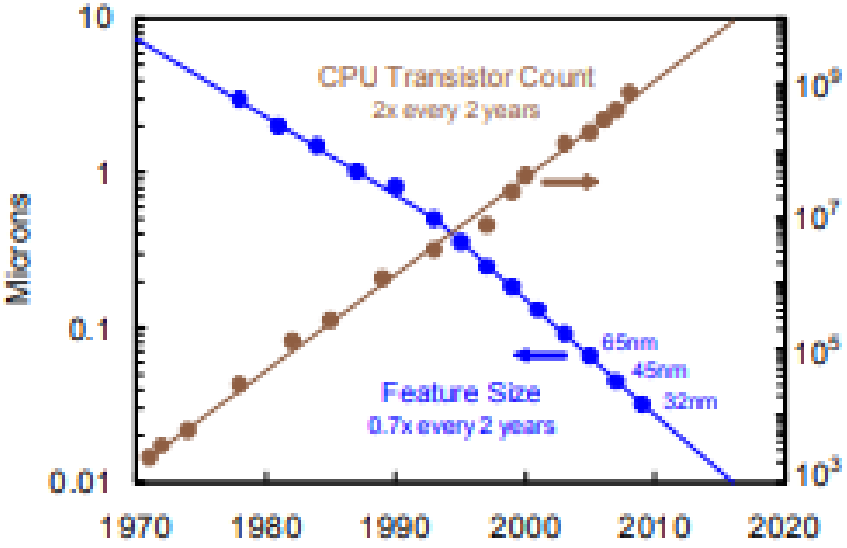
Introduction & Overview

Not restricted to Mechanical Engineering alone

- **Chemical reaction will proceed or not**
- **We grow old and our hair becomes grey**

Why should an electrical engineer be concerned about thermofluids? Cooling!!





10 nm – 2016

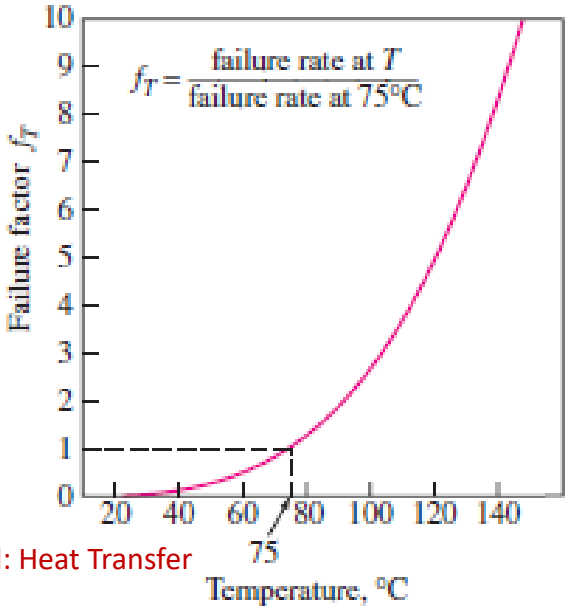
7 nm – 2018

5 nm – 2020

Future

3 nm – ~2022

2 nm – >2023

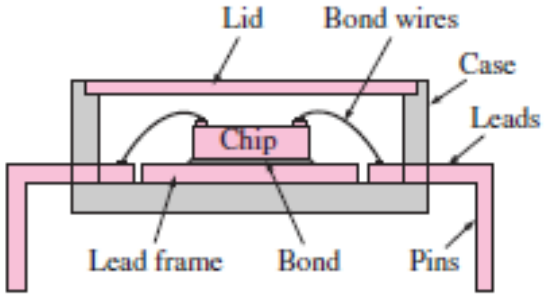


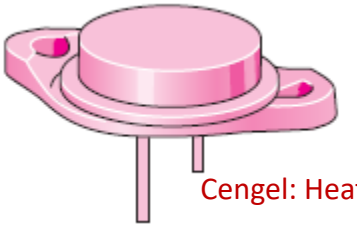
Cengel: Heat Transfer

doi:10.1109/iwce.2009.5091124

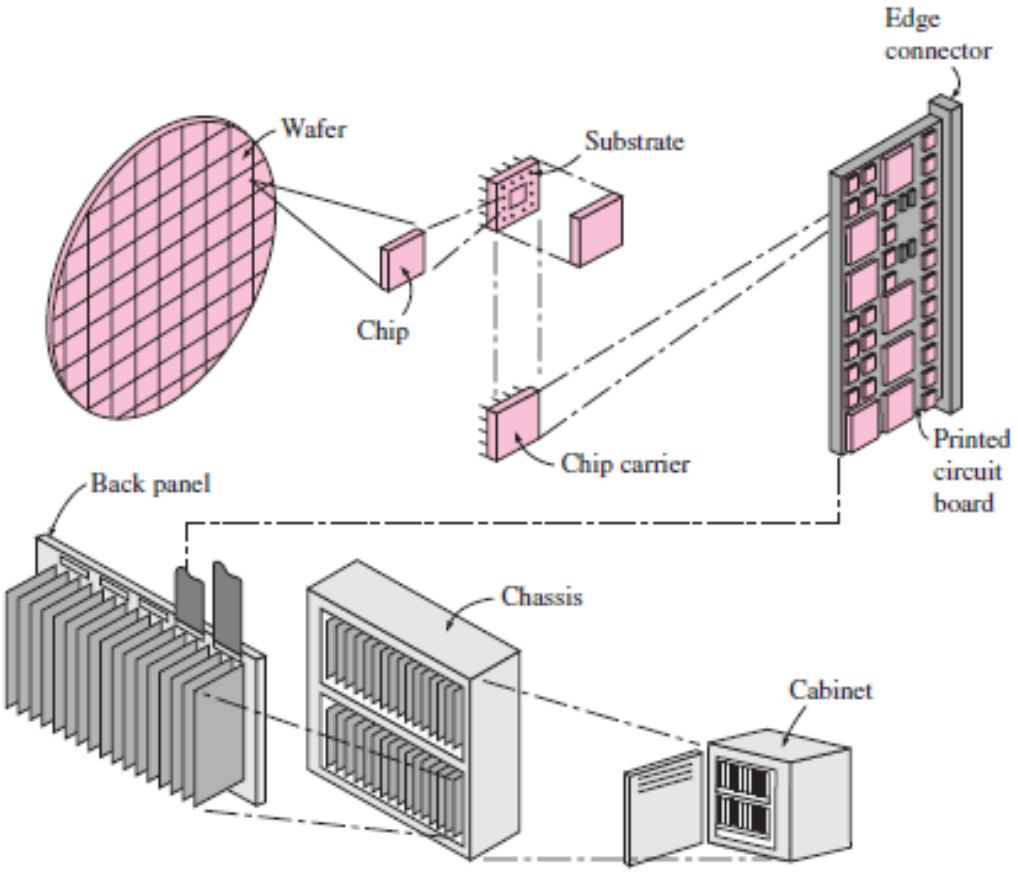
Different applications needing cooling

Example 1: Chip carrier

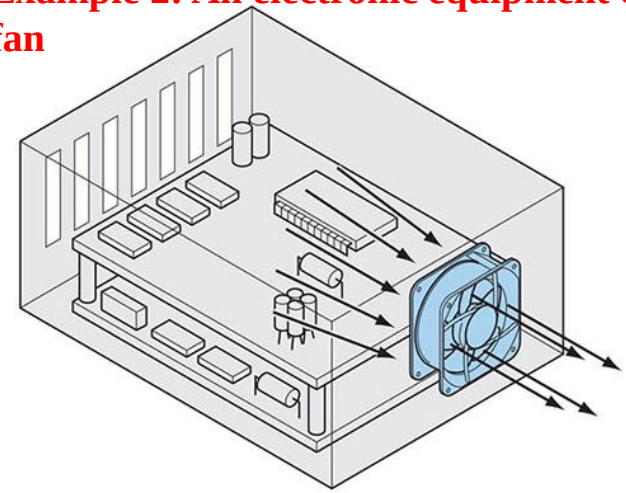




Cengel: Heat Transfer



Example 2: An electronic equipment cooled by fan



Different applications needing cooling

Example 3: Design of a cabinet for cooling electric/electronic equipment

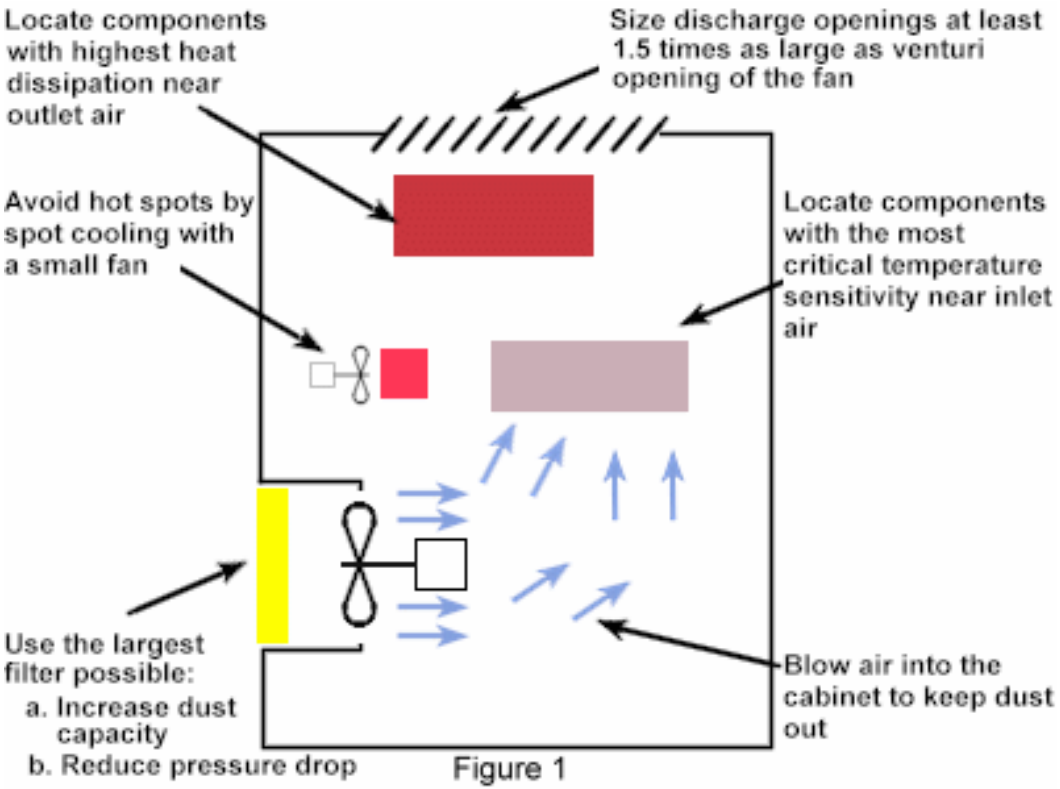
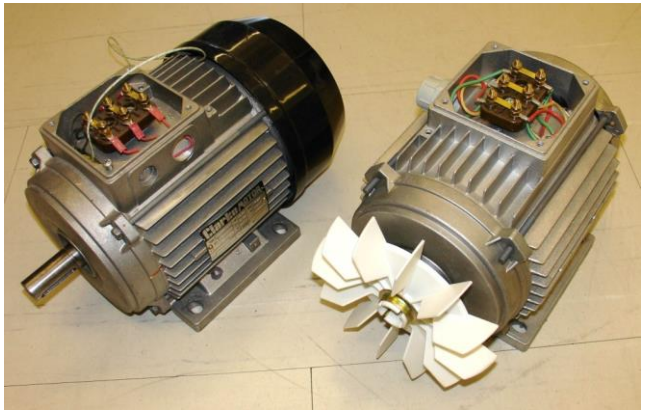
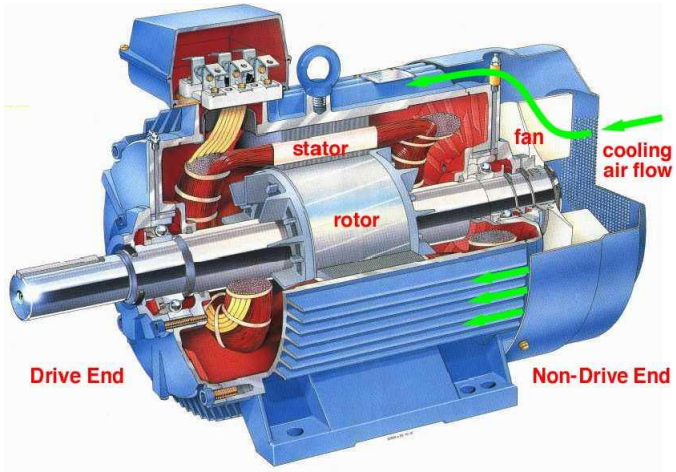


Figure 1

Example 4: An electric motor cooled by a fan

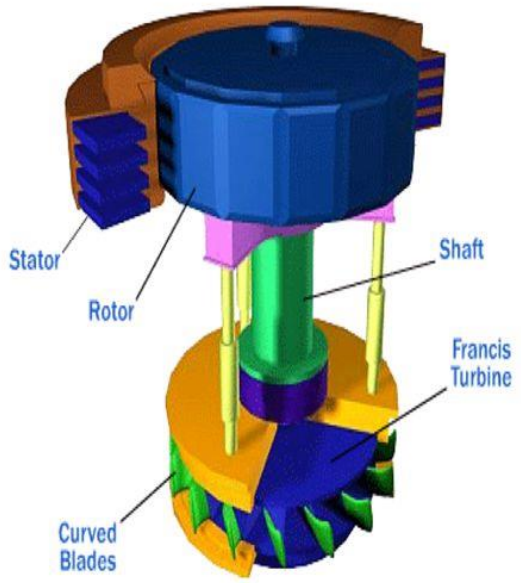


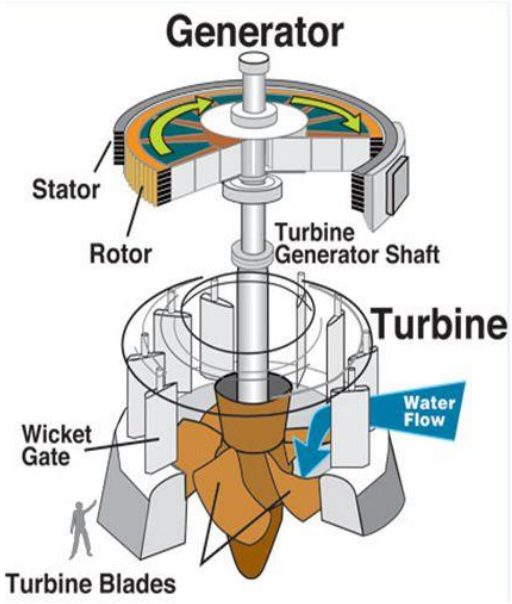


From internet

Why should an electrical engineer be concerned about thermofluids? Not just Cooling!!

Generator – Turbine assembly





From internet

Introduction & Overview

Natural and man-made changes **must** obey the laws of thermodynamics

Law of energy conservation: first law of thermodynamics

Keeping aside the conversion of mass to energy, energy can neither be created nor be destroyed. It can be transformed from one form to another.

Introduction & Overview

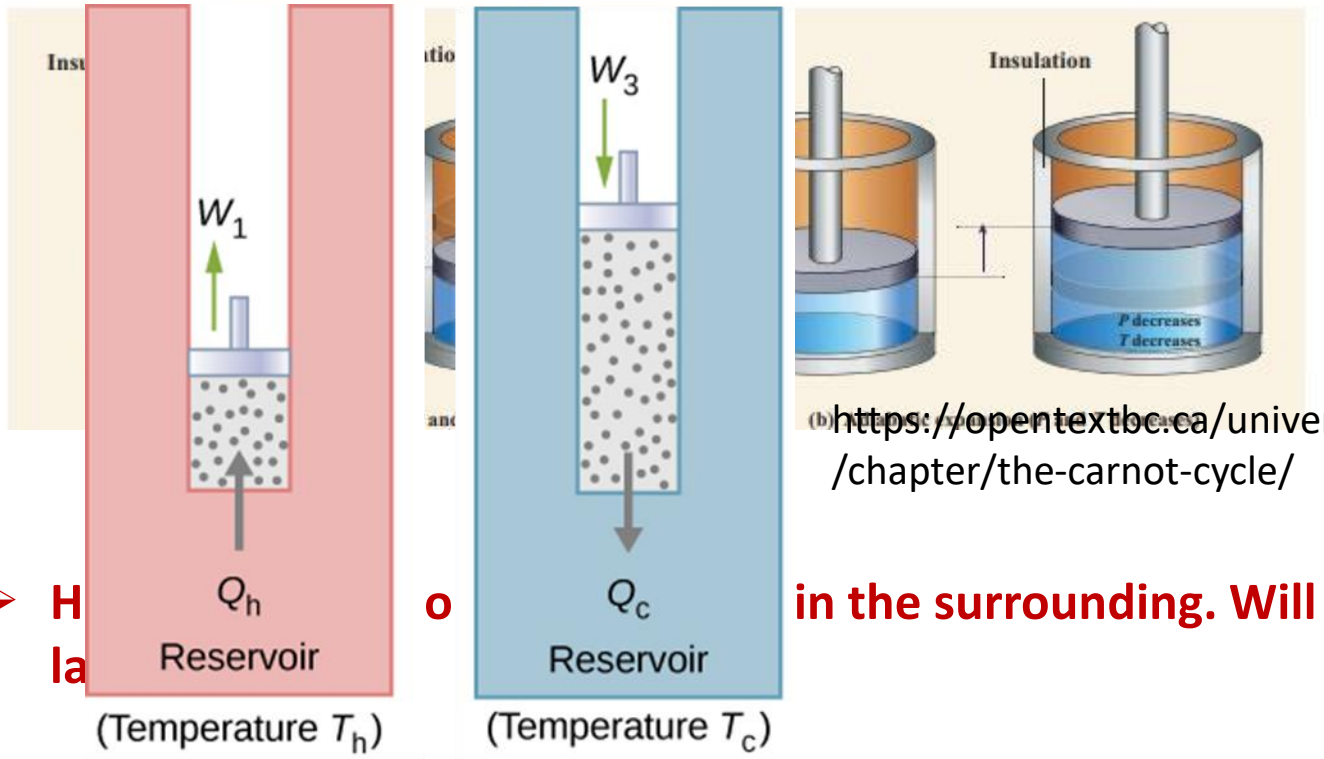
- **Changes do not take place in any direction**
  - **Water flows down from the hill to the valley. Converse?**
  - **Heat flows from high temperature to low temperature. Converse?**

These are examples of natural spontaneous processes

- **Applying brake on a rotating wheel stops motion. Heat is generated.**
- **If the heated wheel is now cooled, will it start rotating? NO**

Introduction & Overview

- **Let us consider the expansion (or, compression) of gas. The gas temperature, pressure or volume may be unchanged after expansion or compression.**



https://www.brainkart.com/article/Adiabatic-process_36250/

https://opentextbc.ca/universityphysicsv2openstax/chapter/the-carnot-cycle/

- **H o in the surrounding. Will talk about it la**

Introduction & Overview

- **Conversion of energy is not efficient in all directions**

  - **Generator/Motor: 100% conversion in ideal case?** **YES**

Kelvin-Plank statement

  - **Heat to work: Is 100% efficiency in conversion possible?** **NO**

Joule's experiment

  - **Work to heat: Is 100% conversion possible?** **YES**

- **Second law: directionality and puts limits on the conversion of energy from one form to another**
- **Thermodynamics deals with physical processes associated with the transfer of energy**
- **The relationship between physical properties is also governed by thermodynamics**

Macroscopic vs Microscopic approach

- **Thermodynamics can be applied to reactions occurring between tiny molecules or to gross changes that occur over the whole universe.**

- **It can be studied using knowledge of molecular-level information: physicists often use this approach**



Statistical thermodynamics

- **It can be studied using averaged physical quantities: engineers often use this approach**

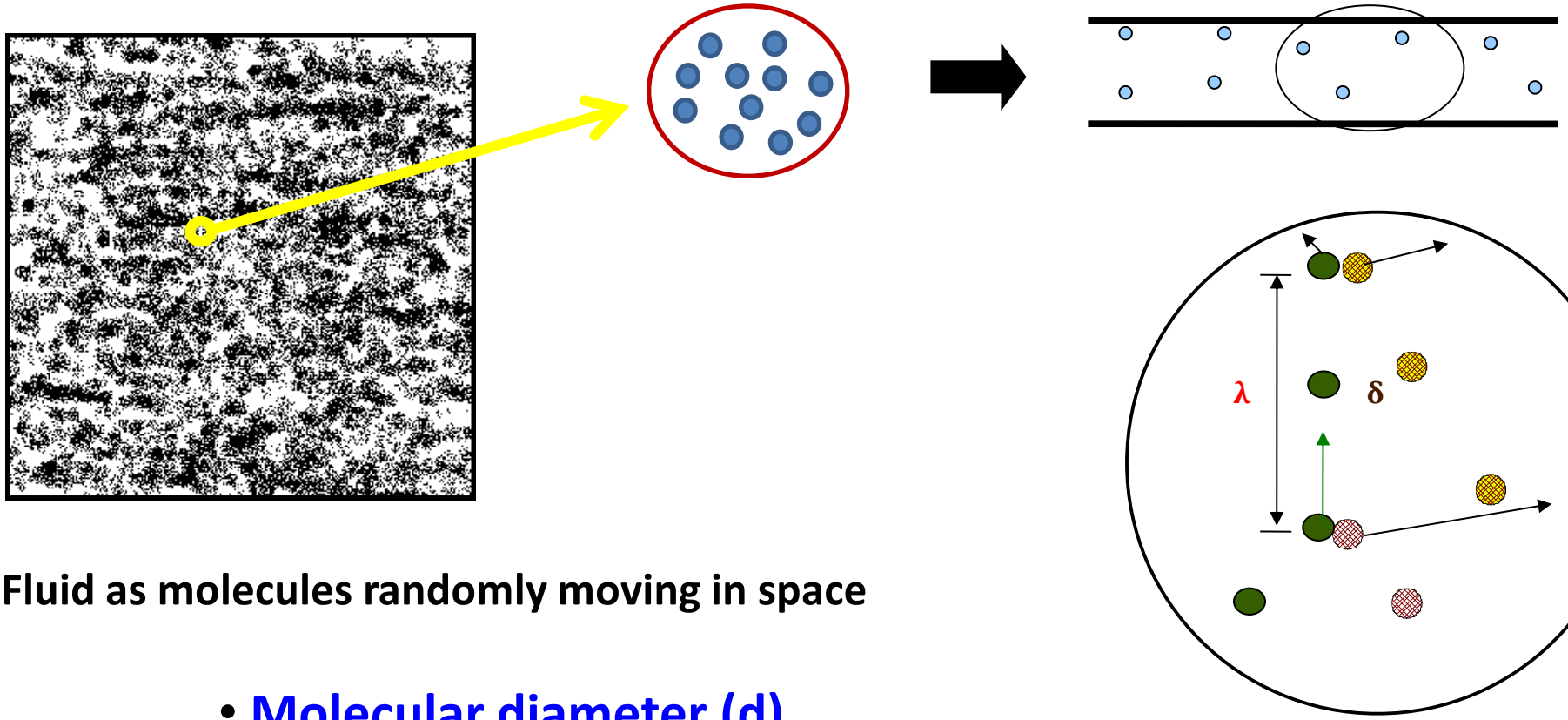


Classical thermodynamics

The macroscopic approach is used in this study

Fluid as Continuum

Microscopic description

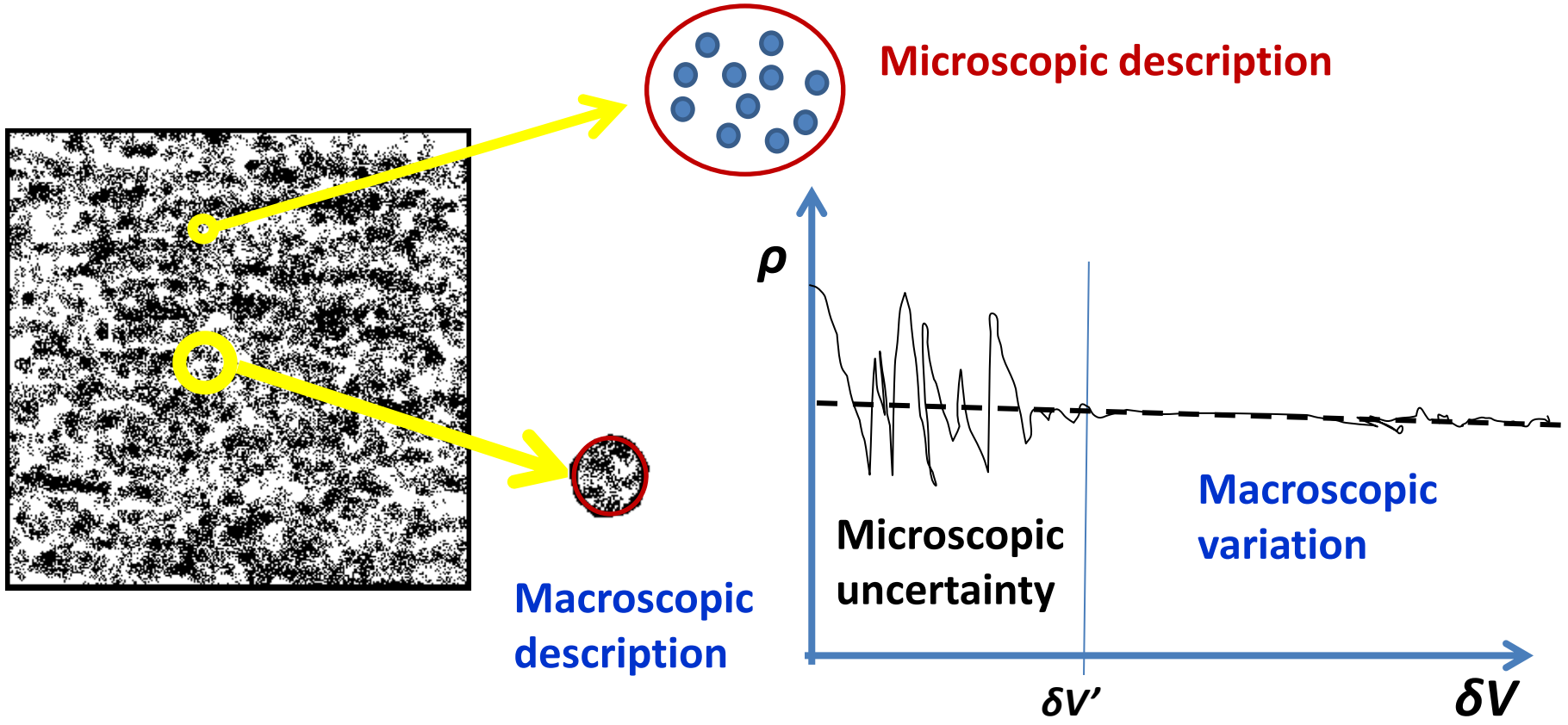


Fluid as molecules randomly moving in space

- **Molecular diameter (d)**
- **Mean molecular spacing (δ)**
- **Mean free path (λ)**

Fluid as Continuum

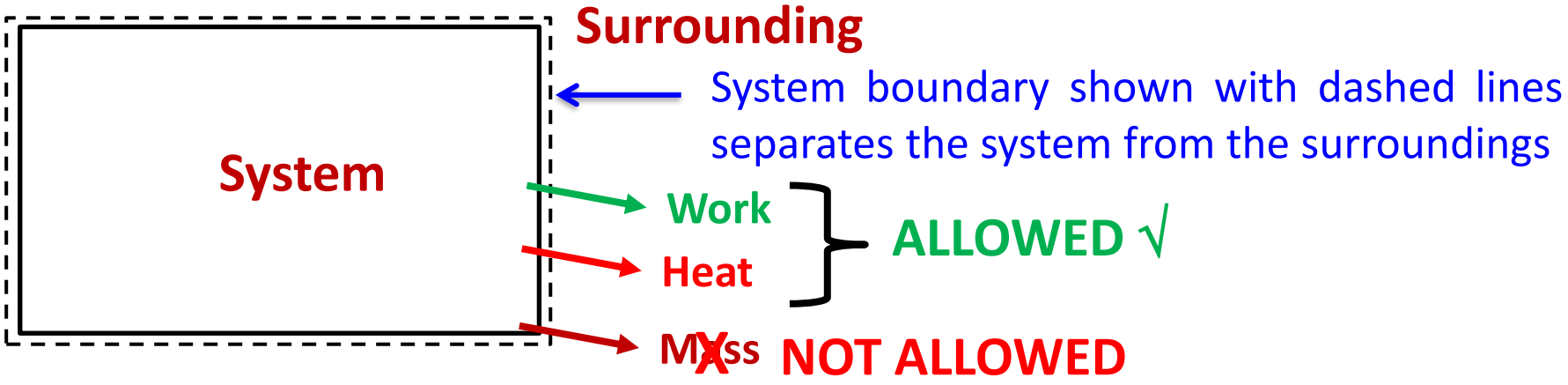
Fluid properties (ρ, etc) can be defined as a continuous field in a continuum



Macroscopic – average effects of several distinguishable constituents (relationships between measurable quantities)

System

A **system** is defined as a **fixed mass and identity** of matter on which attention is focussed. The rest of the universe, which lies outside the system, is called the **surrounding.**



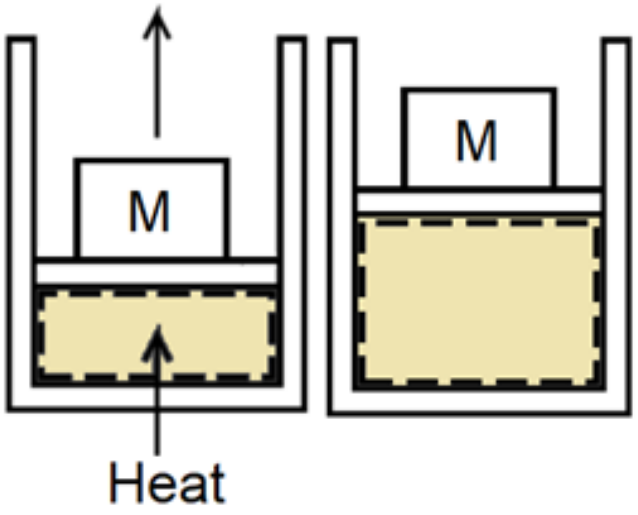
- **System boundary can be a real or an imaginary boundary**
- **The shape and size (volume) of the boundary can change with time**
- **Mass transfer between system and surrounding is NOT allowed**
- **Energy transfer (heat and work interactions) is allowed**

Examples of a system

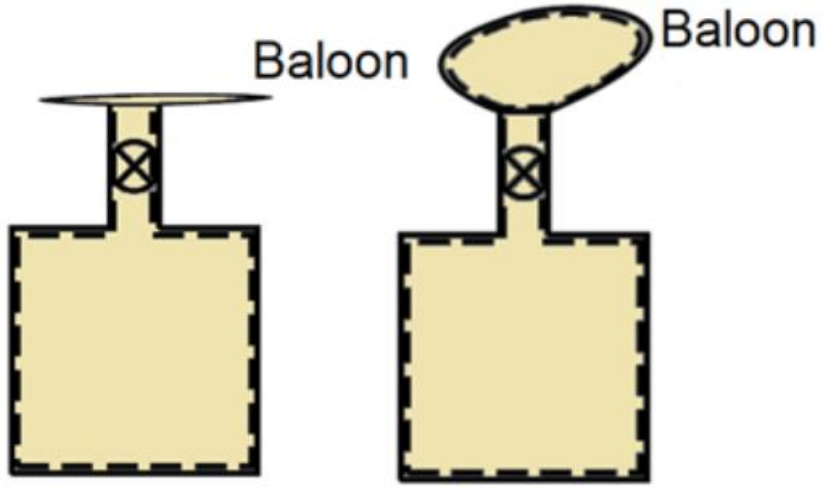
- An electric bulb,
- Gas enclosed in a cylinder-piston device,
- Electrical heater,
- Photovoltaic solar cell,
- Domestic refrigerator (with the door closed),
- a balloon filled with gas, etc.

Are these systems?

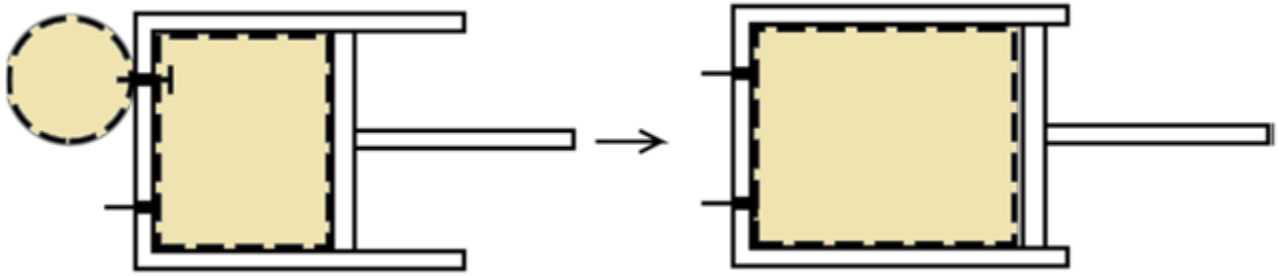
A



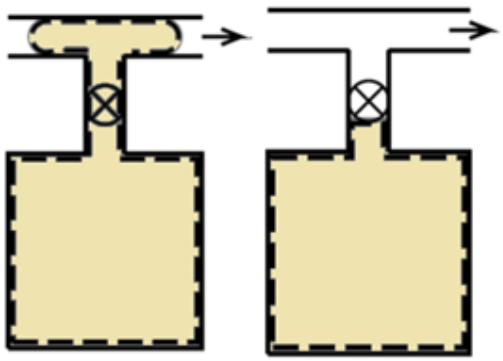
B



C

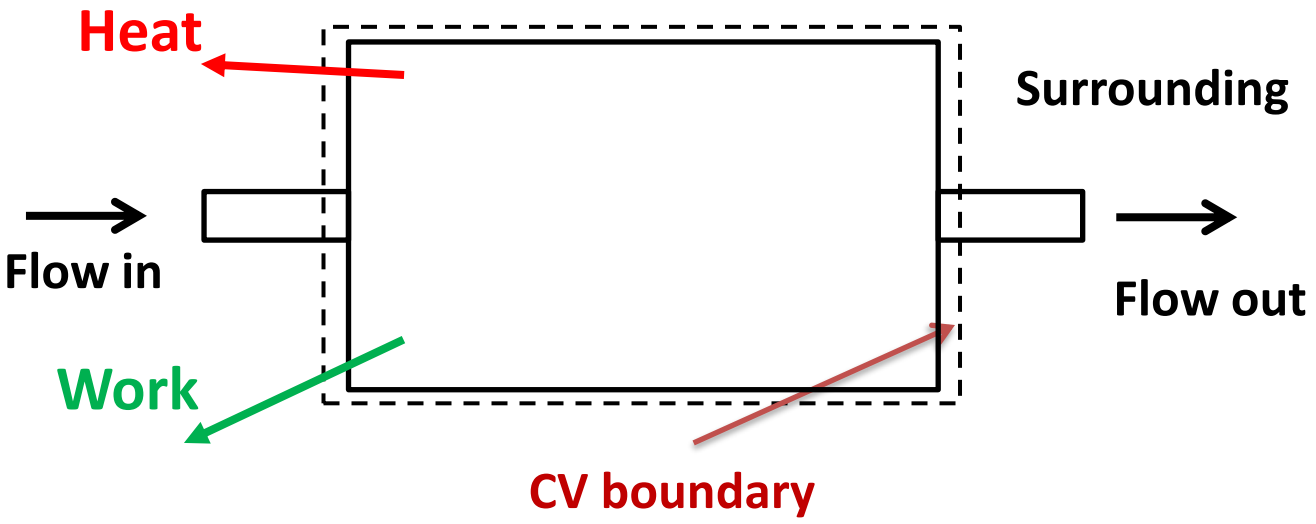


D

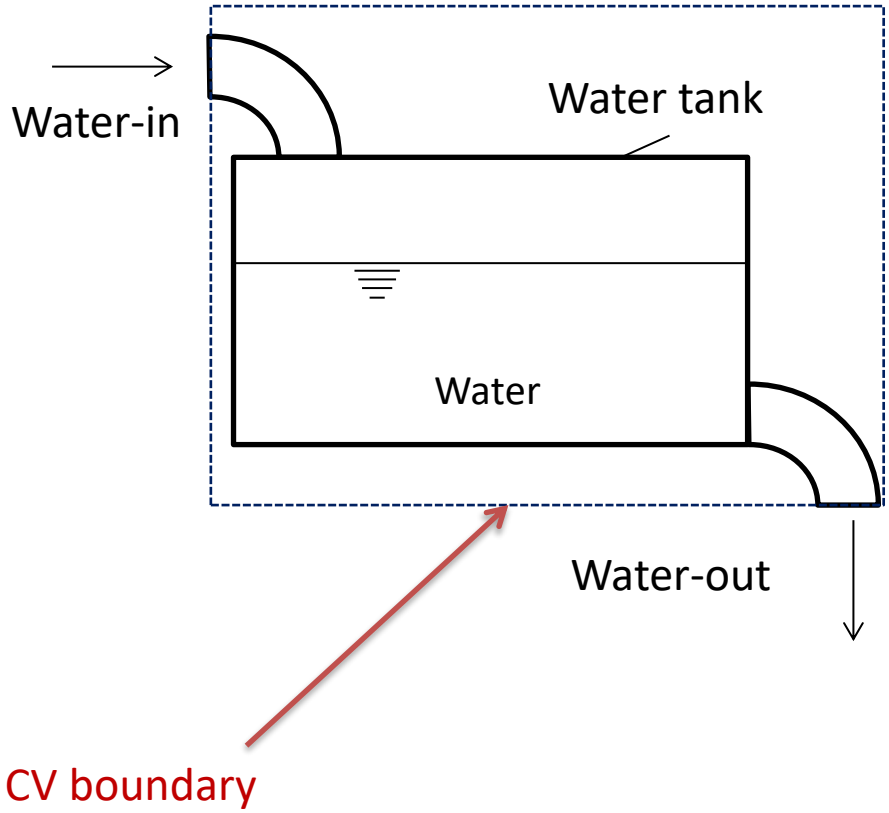


Control volume

- A **control volume** is a fixed volume in space on which attention is focussed
- Energy, **as well as mass transfer**, can occur between the system and surrounding



Control volume - Example



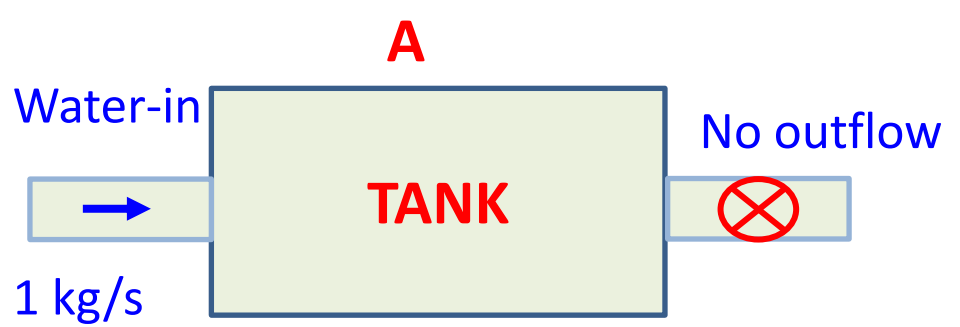
More examples of control volume

- Water pump, air conditioner, car engine, etc.
- Note that both energy and mass transfer can occurs between the control volume and the surroundings in all the above examples

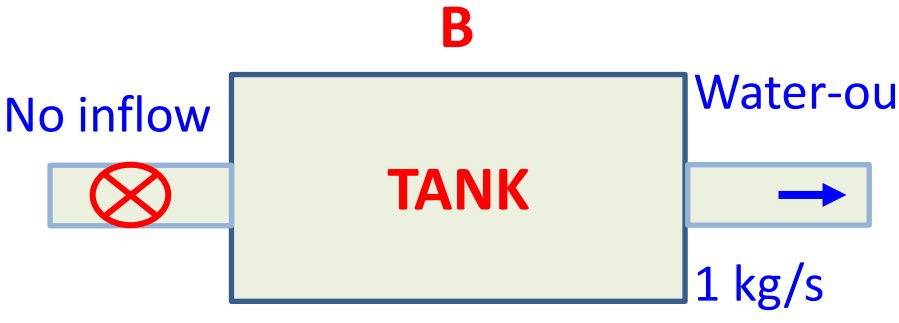
Air is cooled in an air conditioner

Mass transfer between control volume and surroundings ?

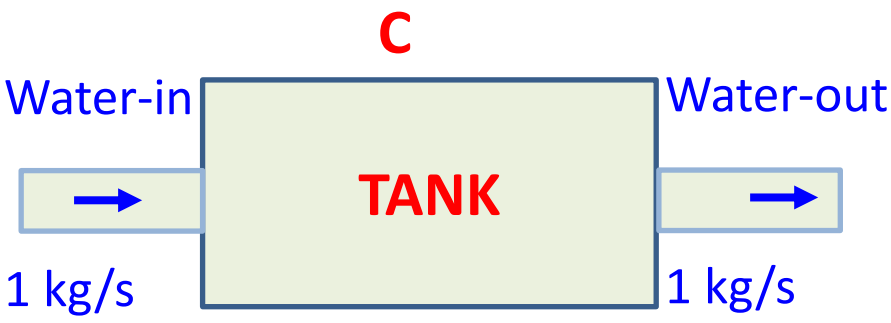
Is it an example of a system or a control volume?



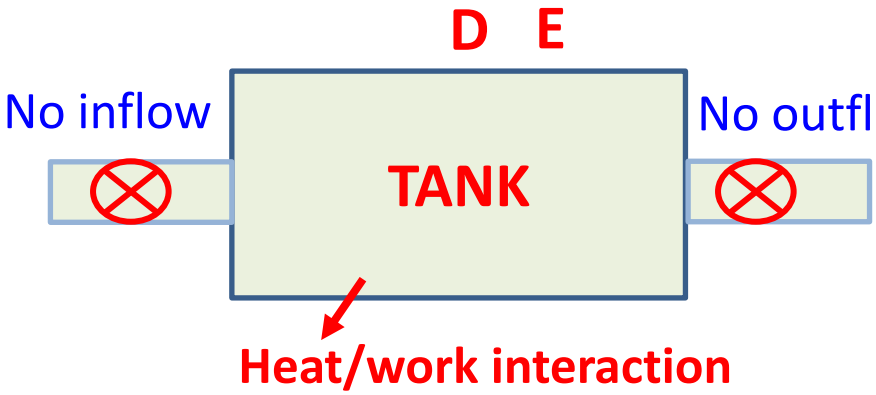
No heat/work interaction



No heat/work interaction



No heat/work interaction



Heat/work interaction

No heat/work interaction

E: ISOLATED SYSTEM

Property, state

Any ***measurable and observable*** **macroscopic characteristic** of a substance is called a property.

Example: Mass, volume, density, pressure, temperature, velocity, elevation, specific heat capacity, kinetic energy, enthalpy, entropy, etc.

Intensive property: **Independent of mass of the system** (pressure, temperature, density)

Extensive property: **Dependent on the size/mass of the system** (volume, potential energy, kinetic energy) **V, E, H, U**

Specific property: **Extensive property per mass**
specific internal energy (J/kg),
specific volume (m^3/kg) **v, e, h, u**
Extensive property per mole (J/mol, m^3/mol)

State

A **state** represents the **collection of all properties of a system** when the system is in **thermodynamic equilibrium**.

- All property values throughout the system should be **uniform**.
- At least for a moment they should be **invariant of time**

A state can be represented as a point in a **thermodynamic coordinate system**. Hence, the property is called a **point function or state variable**.

How many properties are required to define the thermodynamic coordinate system?

For ideal gas: a state diagram is drawn with any two properties (say, pressure and volume) as x and y-axes

Thermodynamic Equilibrium

Thermodynamic equilibrium is ensured by the attainment of **mechanical equilibrium** (no force imbalance between the system and the surroundings), **thermal equilibrium** (no temperature difference between the system and the surroundings) and **chemical equilibrium** (same chemical composition and aggregation).



Let us consider an ideal gas inside an **insulated rigid container**

Gas pressure: 2 bar
Gas temperature: 70 °C

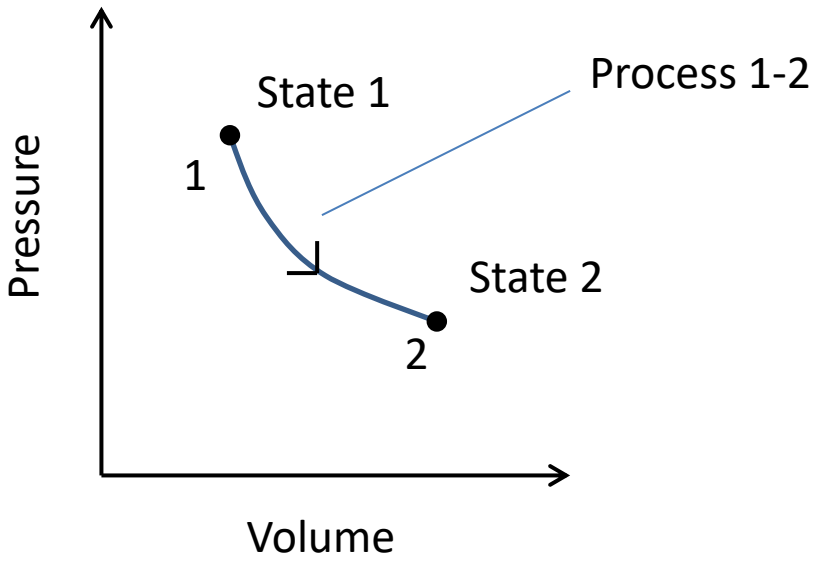
Surrounding has air
Air pressure: 1 bar
Air temperature: 35 °C

Is it interesting to have a strict thermodynamic equilibrium? **NO**

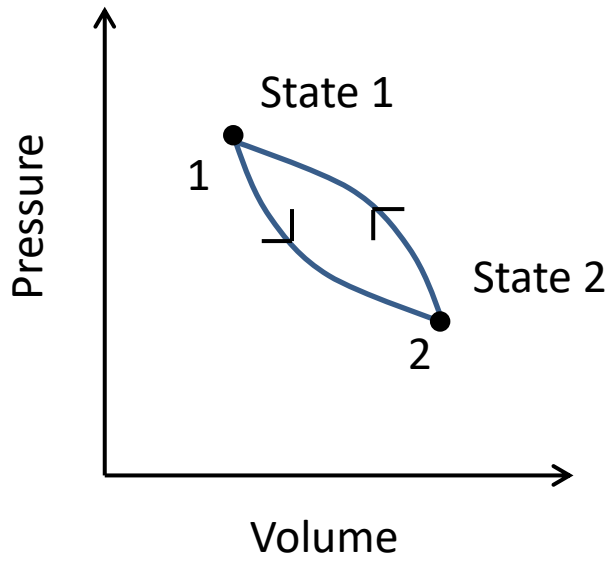
Property, equilibrium, state, process

- **Change of state** occurs when any property of the system changes its value.
- **Path** is the **locus of all the intermediate states** that the system passes through when there is a change of state.
- **Process** is a **sequence of states** by which the change of state takes place in a particular path – mathematical relation
- **Cyclic process** in which the system changes its state more than once and **finally reaches the same initial state** – **there is no net change in any property during a cyclic process.**
- **Quasi-static process** is a slow process in which the intermediate states are well accounted. The process has several **intermediate "equilibrium" states.**
- Properties are defined **only in quasi-static processes**.
- Without equilibrium, different parts of the system would have different properties

State, process and cyclic process



State, Process



Cyclic process

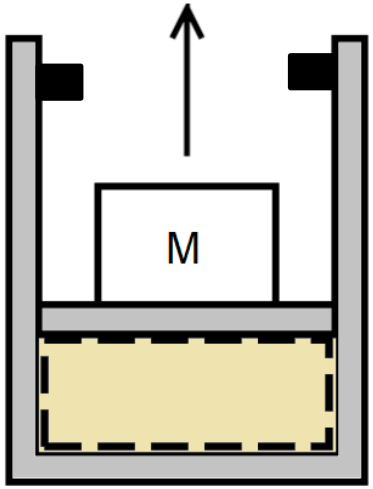
During a **change of state**, the **exact process or path** followed by the system should be known in order to evaluate the **energy transfer in terms of work or heat.**

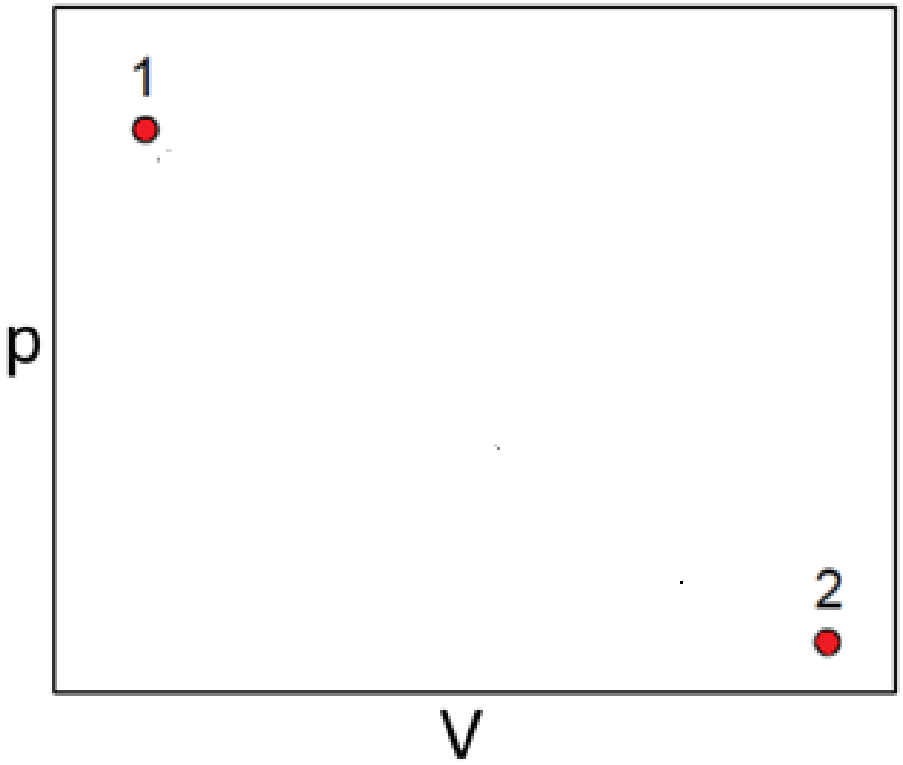
Quasi-static process

What happens when:

Mass M on top of the piston is suddenly removed

Gas volume increases rapidly and reaches a new value





Does this qualify as a quasi-static process?

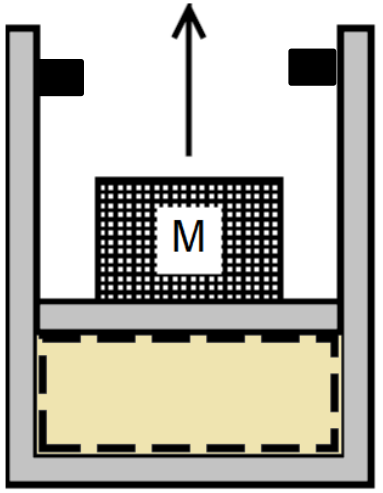
NO

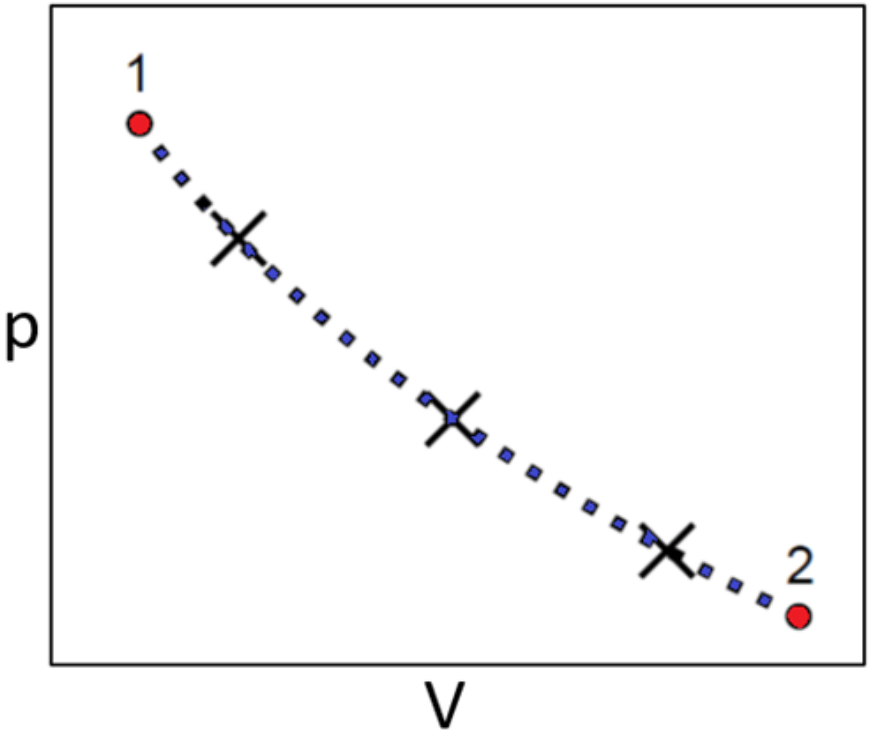
Quasi-static process

What happens when:

Mass M on top of the piston is gradually removed

Gas volume increases slowly and reaches a new value





Does this qualify as a quasi-static process?

YES

Quasi-static Process

- Pressure, volume, and temperature **at each intermediate state** are **measurable.**
- System goes through **several intermediate states** to reach the **same final state 2.**

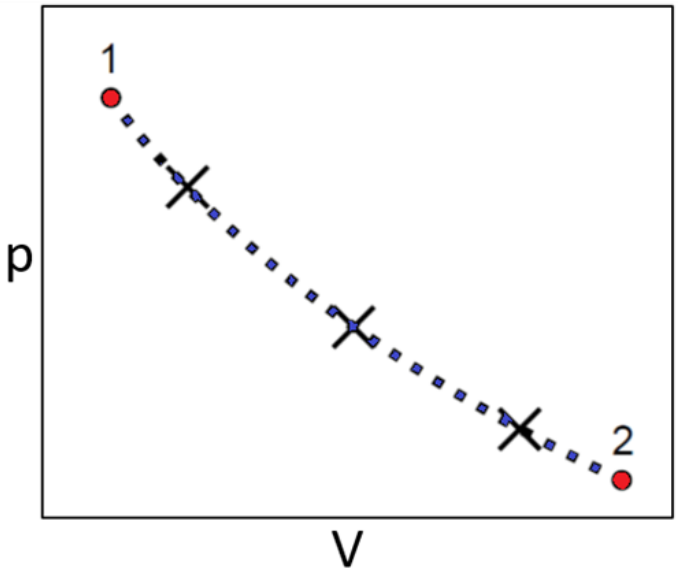
- Number of the known intermediate states increases → the thermodynamic path (also termed as the process) gets defined.

- **Energy transfers can be calculated only when the process is quasi-static.**

- Quasi-static process (also called **quasi-equilibrium process**) may also be a reversible process.
  - If the several **smaller weights are put back one by one**, the system may **follow the same path back to the initial state**, if the **friction between the piston and cylinder is negligible**.

Properties are path independent

- **Properties are unique to a state.**
- The values **are the same at a point (state) irrespective of the thermodynamic path taken** to reach the present state from a previous state.
- They are called **exact differentials**. For example, a change in property is written as ***dΦ***.



$$\int_1^2 d\Phi = \Phi_2 - \Phi_1$$

Exact and Inexact Differentials

Consider a function of two variables $z=f(x,y)$

The total differential of this function is defined as

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy$$

Any differential of this form is exact

In general, a differential dz expressed as $dz = M(x,y)dx + N(x,y)dy$

dz is total differential only if $M(x,y) = \left(\frac{\partial z}{\partial x}\right)_y$ and $N(x,y) = \left(\frac{\partial z}{\partial y}\right)_x$

Condition for exactness: $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ In this case dz is exact and state function

If $M(x,y) \neq \left(\frac{\partial z}{\partial x}\right)_y$ or $N(x,y) \neq \left(\frac{\partial z}{\partial y}\right)_x$

dz is not total differential and it is inexact, z is not a point or state function
And represented as δz

Ideal gas

$$pv = RT$$

Take, volume $v = v(T, p)$

We know that *v* is a property

$$dv = \left(\frac{\partial v}{\partial T}\right)_p dT + \left(\frac{\partial v}{\partial p}\right)_T dp = \frac{R}{p} dT - \frac{RT}{p^2} dp$$

$$\left(\frac{\partial \left(\frac{R}{p}\right)}{\partial p}\right)_T = -R/p^2 \qquad \left(\frac{\partial \left(\frac{RT}{p^2}\right)}{\partial T}\right)_p = -R/p^2$$

Work

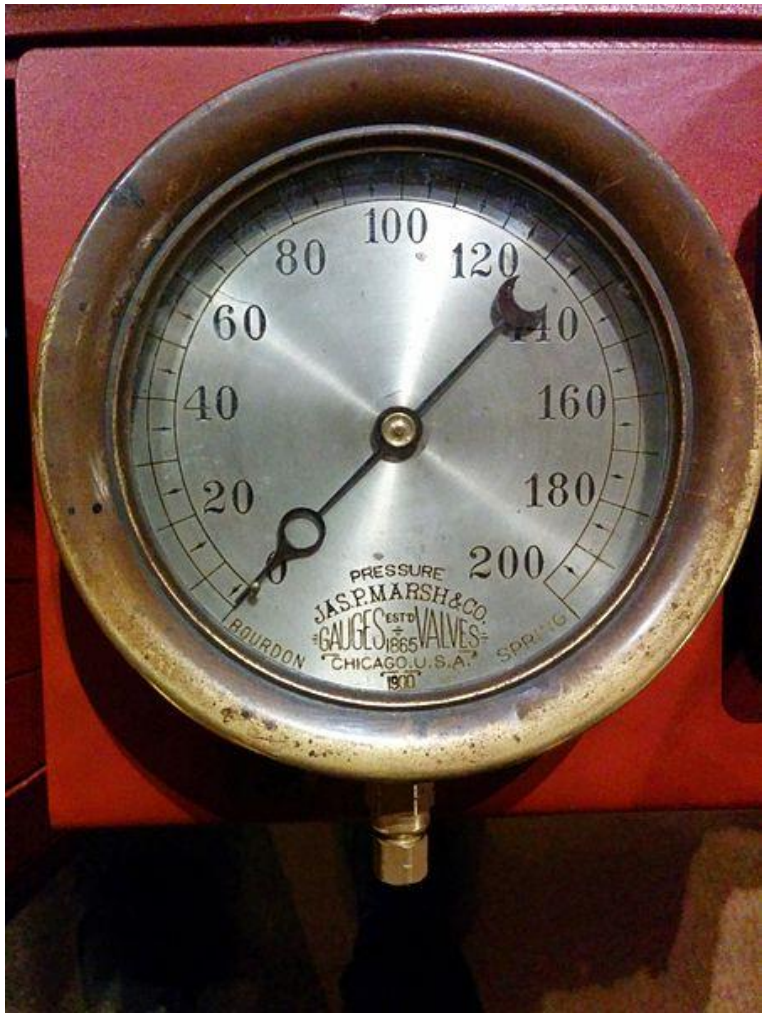
$$dW = pdv = RdT - \frac{RT}{p} dp \qquad \delta W = pdv = RdT - \frac{RT}{p} dp$$

$$\left(\frac{\partial (R)}{\partial p}\right)_T = 0 \qquad \left(\frac{\partial \left(\frac{RT}{p}\right)}{\partial T}\right)_p = \frac{R}{p}$$

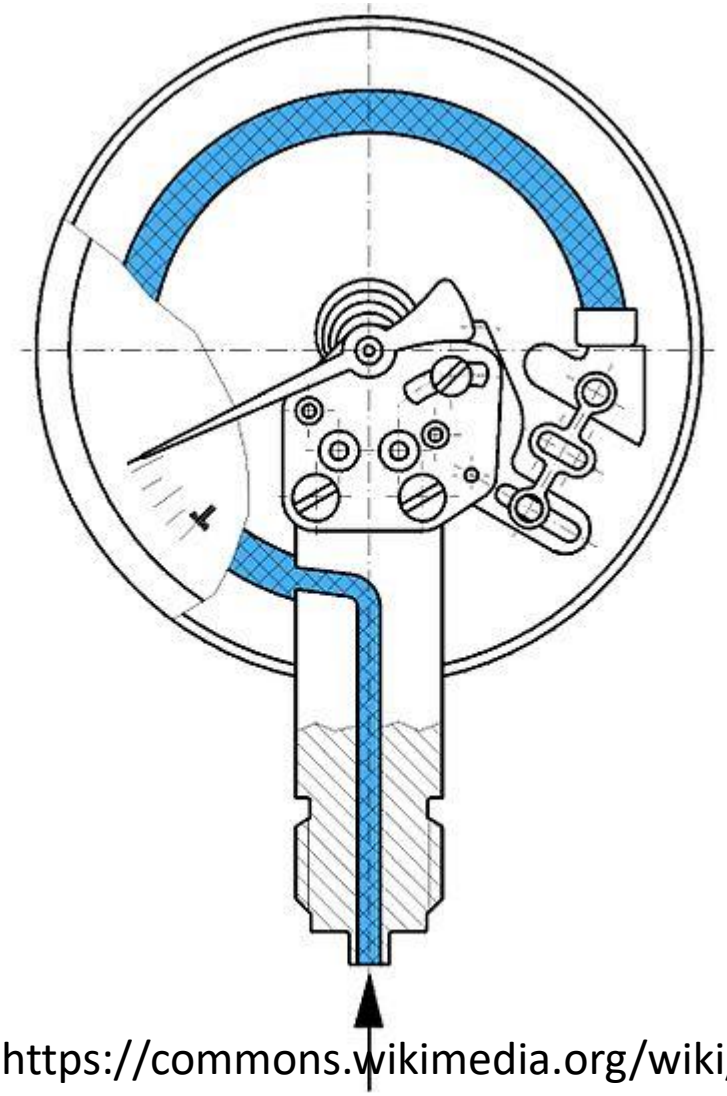
Pressure

- For a **fluid at rest**, the **pressure exerted is the same in all directions**: normal component of the force exerted per unit area (N/m^2) or (Pascal, Pa).
- Commonly used pressure gauges: manometers, Bourdon gauge, barometer
- **Manometer/Bourdon gauge measure pressure w.r.t surrounding pressure**.
- **Barometer measures the absolute pressure**

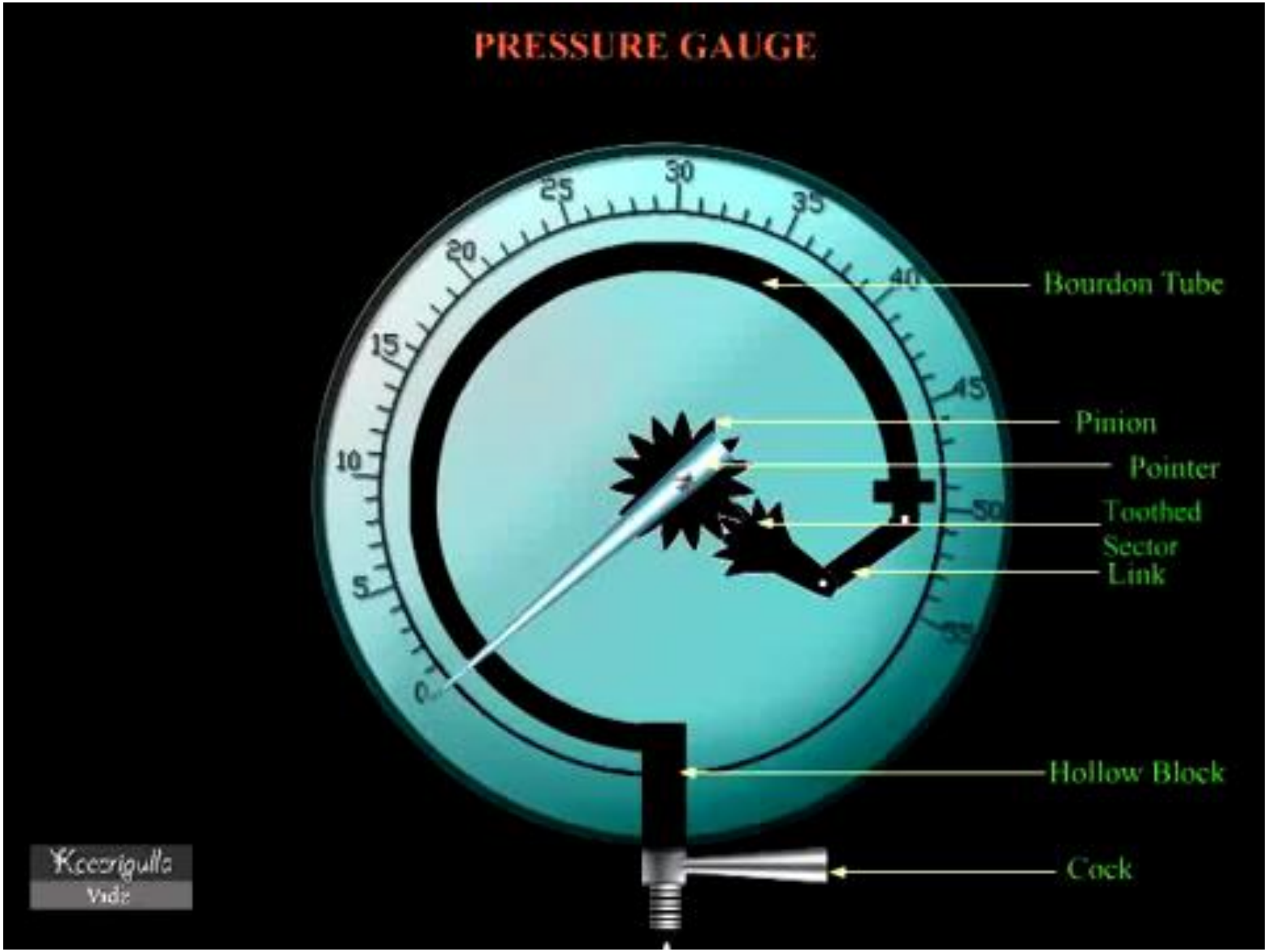
BOURDON GAUGE



https://commons.wikimedia.org/wiki/File:Pressure_gauge_circa_1900.jpg



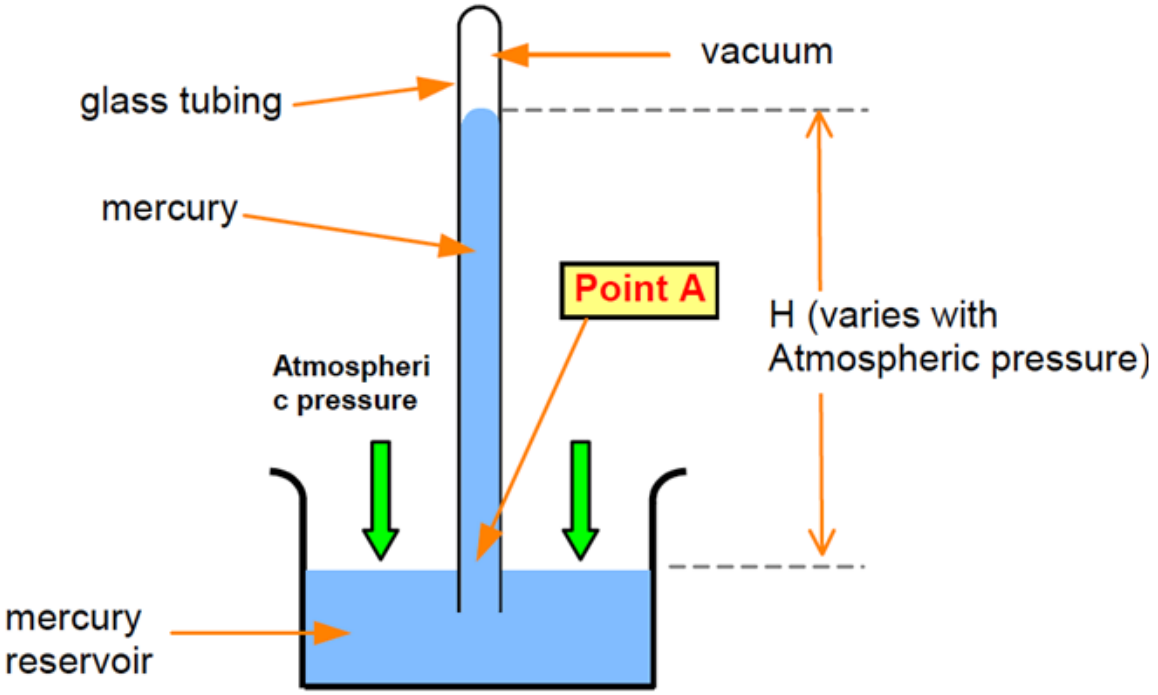
https://commons.wikimedia.org/wiki/File:Rohrfeder_colorida.jpg


PRESSURE GAUGE
Bourdon Tube
Pinion
Pointer
Toothed
Sector
Link
Hollow Block
Cock
Keerigulla
Vade

Mercury Barometer



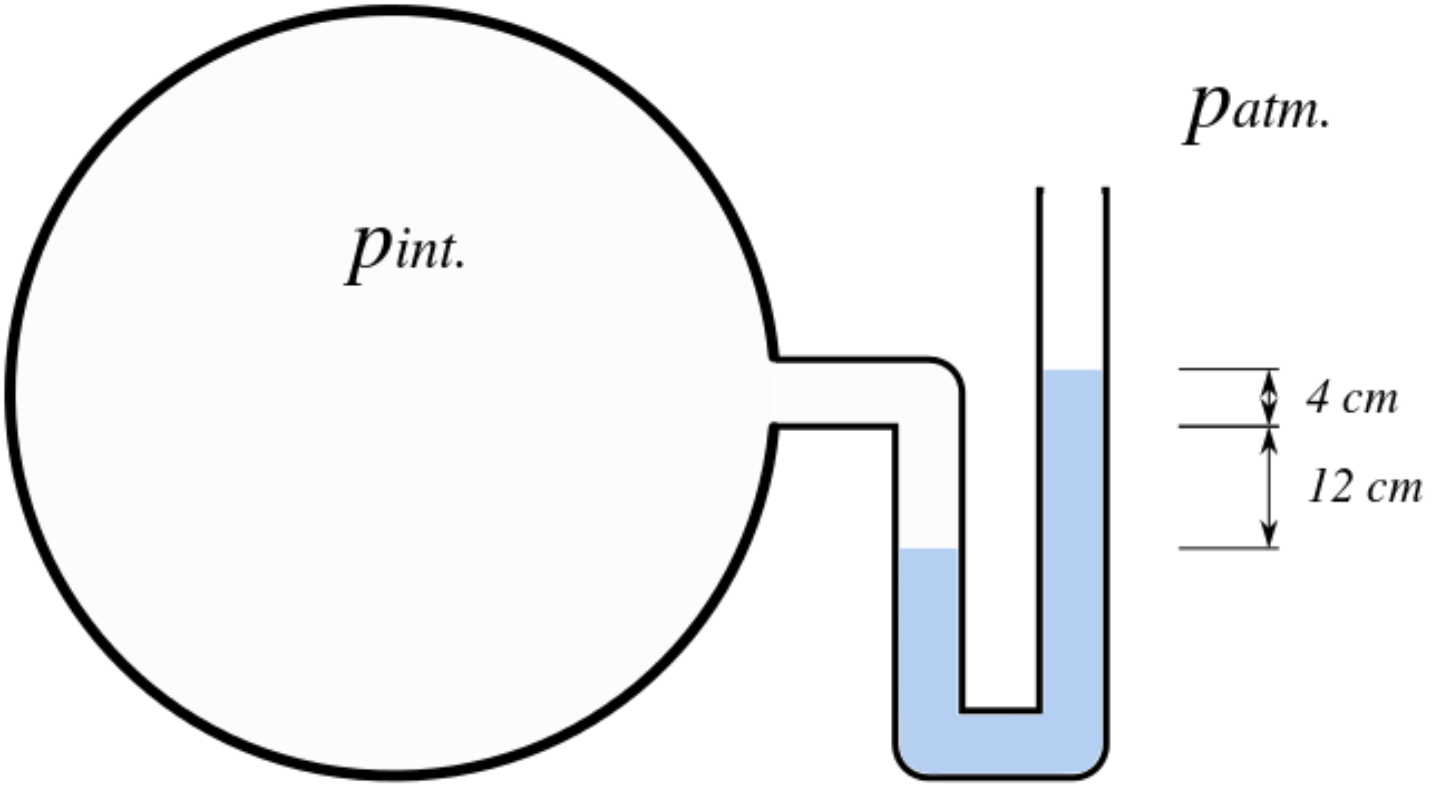
https://www.britannica.com/technology/barometer



Atmospheric pressure = ρgH

https://instrumentationtools.com/what-is-a-barometer/

U-Tube manometer



Thank You